

JEE MAIN REVISION MASTER: ALTERNATING CURRENT

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 191 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	AC Fundamentals: Peak, RMS & Phase	57 PYQs	Foundations
Very High	Reactance: Inductive & Capacitive	57 PYQs	Impedance Elements
Very High	Series LCR Circuit & Impedance	50 PYQs	Circuits
Very High	Resonance, Q-Factor & Bandwidth	40 PYQs	Resonance
Very High	LC Oscillations & Transformers	15 PYQs	Devices & Oscillations
High	Power in AC Circuits & Power Factor	10 PYQs	Power & Efficiency

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
AC Fundamentals: Peak, RMS & Phase	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$, $I_{\text{avg, half}} = \frac{I_0}{2}$
Reactance: Inductive & Capacitive	$X_L = \omega L = 2\pi f L$, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
Series LCR Circuit & Impedance	$Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\phi = \frac{X_L - X_C}{R}$, $V_{\text{total}} = \sqrt{V_R^2 + (V_L - V_C)^2}$
Resonance, Q-Factor & Bandwidth	$\omega_0 = \frac{1}{\sqrt{LC}}$, $Z_{\min} = R$, $I_{\max} = \frac{V}{R}$, $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0}{\Delta\omega}$
LC Oscillations & Transformers	$V_s V_p = N_s N_p = I_p I_s$, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$, $q(t) = q_0 \cos(\omega_0 t)$
Power in AC Circuits & Power Factor	$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi$, $\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$, $I_{\text{wattless}} = I_{\text{rms}} \sin\phi$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 AC Fundamentals: Peak, RMS & Phase (Very High Yield — 57 PYQs)

Core Principle: Sinusoidal AC quantities: $I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$. Average current over full cycle is 0; average over half cycle is $I_{\text{avg, half}} = (2I_0 / \pi) \approx 0.637 I_0$.

■ **Caution / JEE Pitfall:** AC voltmeters and ammeters measure RMS values, not peak values! When JEE specifies '220 V AC', it always denotes V_{rms} (peak voltage is $V_0 = 220\sqrt{2} \approx 311 \text{ V}$).

■ **Speed Trick:** Instantaneous power $p(t) = v(t)i(t)$ fluctuates at double frequency (2ω) about average power $P_{\text{avg}} = V_{\text{rms}}I_{\text{rms}}\cos\phi$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#3 (Numerical):** "An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (r.m.s.) is connected to it." **[Ans: 314]**
- **[2026 Morning Shift] Q#4 (MCQ):** "The electric current in the circuit is given as $i = i_0 \sin(\omega t)$. The r.m.s current for the period $t=0$ to $t=T$ is _____." **[Ans: B]**
- **[2025 Evening Shift] Q#1 (MCQ):** "An electric bulb rated as 100W-220V is connected to an ac source of rms voltage 220 V. The peak value of current through the bulb is _____." **[Ans: N/A]**

3.2 Reactance: Inductive & Capacitive (Very High Yield — 57 PYQs)

Core Principle: Pure Inductor: Voltage leads current by 90° ($\pi/2$), inductive reactance $X_L = \omega L = 2\pi f L$. Pure Capacitor: Current leads voltage by 90° , capacitive reactance $X_C = (1 / \omega C) = (1 / 2\pi f C)$.

■ **Caution / JEE Pitfall:** In direct current (DC, $f = 0$), an inductor offers zero reactance ($X_L = 0$, short circuit) while a capacitor offers infinite reactance ($X_C \rightarrow \infty$, blocking DC completely).

■ **Speed Trick:** Mnemonic 'CIVIL': In a Capacitor (C), Current (I) leads Voltage (V); in an Inductor (L), Voltage (V) leads Current (I).

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A LCR series circuit driven with $E_{\text{rms}}=90\text{V}$ at frequency $f=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactance $X_L=60\Omega$ and a capacitor with capacitive reactance $X_C=40\Omega$. The average power dissipated in the circuit is _____." **[Ans: A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "For the series LCR circuit connected with 220V, 50Hz a.c source as shown in the figure, the power factor is $\cos\phi$. The value of ϕ is _____." **[Ans: C]**
- **[2026 Evening Shift] Q#1 (Numerical):** "A series LCR circuit with $R=20\Omega$, $L=1.6\text{H}$ and $C=40\mu\text{F}$ is connected to a variable frequency a.c. source. The inductive reactance at resonance is _____." **[Ans: 200]**

3.3 Series LCR Circuit & Impedance (Very High Yield — 50 PYQs)

Core Principle: Series LCR impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Circuit current $I = (V / Z)$. Phase angle $\tan\phi = (X_L - X_C / R)$. Net voltage phasor $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Caution / JEE Pitfall:** Voltages across series AC elements do NOT add algebraically: $V_{\text{(total)}} \neq V_R + V_L + V_C$! Always use phasor vector addition: $V_{\text{(total)}} = \sqrt{V_R^2 + (V_L - V_C)^2}$.

■ **Speed Trick:** If $X_L > X_C$, circuit is inductive (voltage leads current by ϕ). If $X_C > X_L$, circuit is capacitive (current leads voltage by ϕ).

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "A LCR series circuit driven with $E_{\text{(rms)}}=90\text{V}$ at frequency $f_{\text{(d)}}=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactanc..." **[Ans: A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "For the series LCR circuit connected with $220\text{V}, 50\text{Hz}$ a.c source as shown in the figure, the power factor is $(\alpha / 10)$. The value..." **[Ans: C]**
- **[2026 Morning Shift] Q#2 (Numerical):** "An inductor of 10 mH , capacitor of $0.1\mu\text{ F}$ and a resistor of 100Ω are connected in series across an a.c power supply $220\text{V}, 70\text{Hz}$..." **[Ans: 100]**

3.4 Resonance, Q-Factor & Bandwidth (Very High Yield — 40 PYQs)

Core Principle: At resonance: $X_L = X_C$, resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ ($f_0 = \frac{1}{2\pi\sqrt{LC}}$). Minimum impedance $Z_{\text{(min)}} = R$, maximum current $I_{\text{(max)}} = (V / R)$. Quality factor $Q = (\omega_0 L / R) = (1 / \omega_0 C R) = (1 / R) \sqrt{(L / C)} = (\omega_0 / \Delta\omega)$.

■ **Caution / JEE Pitfall:** At resonance, V_L and V_C are NOT zero! In fact, $V_L = V_C = Q \cdot V_{\text{(supply)}}$, which can be substantially higher than supply voltage (voltage magnification)!

■ **Speed Trick:** Resonance bandwidth $\Delta\omega = (R / L)$. To sharpen resonance (higher Q), decrease resistance R or increase the (L / C) ratio.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (Numerical):** "A series LCR circuit with $R=20\Omega, L=1.6\text{H}$ and $C=40\mu\text{ F}$ is connected to a variable frequency a.c. source. The inductive reactance at $r...$ " **[Ans: 200]**
- **[2026 Evening Shift] Q#2 (MCQ):** "A capacitor C is first charged fully with potential difference of $V_{\text{(0)}}$ and disconnected from the battery. The charged capacitor is connec..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#1 (MCQ):** "A series LCR circuit is connected to an alternating source of emf E . The current amplitude at resonant frequency is $I_{\text{(0)}}$. If the value o..." **[Ans: A]**

3.5 LC Oscillations & Transformers (Very High Yield — 15 PYQs)

Core Principle: Ideal transformer equations: $(V_s / V_p) = (N_s / N_p) = (I_p / I_s)$. Efficiency $\eta = \frac{P_{out}}{P_{in}} = (V_s I_s / V_p I_p)$. LC tank oscillations: charge $q(t) = q_0 \cos(\omega_0 t)$ with total energy conserved $U = (q^2 / 2C) + (1 / 2) L I^2$.

■ **Caution / JEE Pitfall:** Transformers operate exclusively on AC; in DC, primary flux is static ($(d\Phi / dt) = 0$), producing zero secondary voltage and potentially burning the primary coil!

■ **Speed Trick:** In step-up transformers ($N_s > N_p$), voltage increases but current decreases, keeping total power $V I$ conserved under ideal conditions.

Representative Source PYQs:

- **[2024 Morning Shift] Q#2 (MCQ):** "Primary coil of a transformer is connected to 220V ac. Primary and secondary turns of the transforms are 100 and 10 respectively. Secondar..." **[Ans: N/A]**
- **[2024 Evening Shift] Q#2 (MCQ):** "Primary side of a transformer is connected to 230V, 50Hz supply. Turns ratio of primary to secondary winding is 10:1. Load resistance co..." **[Ans: N/A]**
- **[2024 Evening Shift] Q#4 (Numerical):** "A power transmission line feeds input power at 2.3kV to a step down transformer with its primary winding having 3000 turns. The output pow..." **[Ans: 45]**

3.6 Power in AC Circuits & Power Factor (High Yield — 10 PYQs)

Core Principle: Average power dissipated $P_{avg} = V_{rms} I_{rms} \cos\phi$. Power factor $\cos\phi = (R / Z) = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$. Wattless current component $I_{wattless} = I_{rms} \sin\phi$.

■ **Caution / JEE Pitfall:** Power factor is $\cos\phi = (R / Z)$, NOT $\tan\phi$ or (R / X) ! For pure L or pure C, $\phi = 90^\circ$ implies $\cos\phi = 0$, so average power dissipation is strictly zero.

■ **Speed Trick:** Apparent power $S = V_{rms} I_{rms}$ (in VA). True power $P = S \cos\phi$ (in W). Reactive power $Q = S \sin\phi$ (in VAR).

Representative Source PYQs:

- **[2025 Evening Shift] Q#1 (Numerical):** "An inductor of reactance 100Ω , a capacitor of reactance 50Ω , and a resistor of resistance 50Ω are connected in series..." **[Ans: 1]**
- **[2024 Evening Shift] Q#1 (Numerical):** "A series LCR circuit with $L = (100 / \pi) \text{ mH}$, $C = \frac{10^{-3}}{3\pi} \text{ F}$ and $R = 10\Omega$, is connected across an ac source of 220V, 50Hz su..." **[Ans: 1]**
- **[2024 Evening Shift] Q#4 (MCQ):** "An AC voltage $V = 20 \sin 200\pi t$ is applied to a series LCR circuit which drives a current $I = 10 \sin(200\pi t + (\pi / 3))$. The average po..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A LCR series circuit driven with $E_{\text{rms}}=90\text{V}$ at frequency $f=30\text{Hz}$ has resistance $R=80\Omega$, an inductance with inductive reactance $X_L=20.0\Omega$ and capacitance with capacitive reactance $X_C=80.0\Omega$. The power factor of the circuit is ____.

- (A) 0.8 [Correct]
- (B) 0.64
- (C) 0.9
- (D) 0.5

Correct Answer: A

Solution Analysis: In an AC circuit, the power factor ($\cos\phi$) is defined as the ratio of the resistance (R) to the total impedance (Z) of the circuit : $\text{Power Factor}=\cos\phi=(R/Z)$ For a series LCR circuit, the impedance is $Z=\sqrt{R^2+(X_C-X_L)^2}$ The value of resistance is $R=80\Omega$, the inductive reactance is $X_L=20.0\Omega$, and the capacitive reactance is $X_C=80.0\Omega$. $X_{\text{net}}=|X_L-X_C|=|20-80|=60\Omega$ So, the impedance is, $Z=\sqrt{80^2+60^2}=\sqrt{6400+3600}=\sqrt{10000}=100\Omega$ Thus, the power factor is, $\cos\phi=\frac{R}{Z}=\frac{80}{100}$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#3)

The figure given below shows an LCR series circuit with two switches S_1 and S_2 . When switch S_1 is closed keeping S_2 open, the phase difference (ϕ) between the current and source voltage is 30° and phase difference is 60° when S_2 is closed keeping S_1 open. The value of $(3L_1 - L_2)$ is _____ H.

- (A) $(9/2)$
- (B) $(2/9)$ [Correct]
- (C) $(1/3)$
- (D) 3

Correct Answer: B

Solution Analysis: In an LCR circuit the phase difference ϕ is given by the formula : $\tan\phi=\frac{X_L-X_C}{R}$ Case 1 : S_1 closed, S_2 open In this case the inductor L_2 is short-circuited. The active components are R, C , and L_1 . $\tan 30^\circ=\frac{X_{L1}-X_C}{R} \Rightarrow \frac{1}{\sqrt{3}}=\frac{\omega L_1-(1/\omega C)}{R} \Rightarrow R=\sqrt{3}(\omega L_1-(1/\omega C))$(i) Case 2 : S_2 closed, S_1 open Here, the inductor L_1 is short-circuited. The active components are R, C , and L_2 . $\tan 60^\circ=\frac{X_{L2}-X_C}{R} \Rightarrow \sqrt{3}=\frac{\omega L_2-(1/\omega C)}{R}$(ii)

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Evening Shift (Q#3)

An inductor stores 16 J of magnetic field energy and dissipates 32 W of thermal energy due to its resistance when an a.c. current of 2 A (rms) and frequency 50 Hz flows through it. The ratio of inductive reactance to its resistance is _____. ($\pi=3.14$)

Correct Answer: 314

Solution Analysis: Energy stored (U) in inductor is $U=16\text{J}$ Power dissipated in resistance is $P=32\text{W}$ RMS current in the circuit is $I_{\text{rms}}=2\text{A}$ Frequency (f) of the ac source is $f=50\text{Hz}$ Power is dissipated only through the resistive component of the inductor. $P=I_{\text{rms}}^2 \cdot R \Rightarrow 32=(2)^2 \cdot R \Rightarrow 32=4R \Rightarrow R=8\Omega$ Magnetic energy is stored in the inductor. $U=(1/2)L \cdot i^2$ Where L is the self inductance of the inductor and i is the current through it. As the current source is AC, so energy stored is equivalent to current corresponding to the DC source

Benchmark Problem 4: JEE Main 2025 (Online) 22nd January Evening Shift (Q#1)

A series LCR circuit is connected to an alternating source of emf E . The current amplitude at resonant frequency is I_0 . If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

- (A) $\frac{I_0}{2}$ [Correct]
- (B) $\frac{I_0}{\sqrt{2}}$
- (C) $2I_0$
- (D) I_0

Correct Answer: A

Solution Analysis: At resonance in a series LCR circuit, the reactive components (inductive and capacitive) cancel each other out, leaving only the resistance. Therefore, the amplitude of the current is given by: $I_0 = (E / R)$ Now, if the resistance is doubled, the new resistance becomes $2R$. The current amplitude at resonance then becomes: $I = (E / 2R) = (1 / 2)(E / R) = \frac{I_0}{2}$ Thus, the amplitude of current at resonance when the resistance is doubled is $\frac{I_0}{2}$. The correct option is A.

Benchmark Problem 5: JEE Main 2025 (Online) 7th April Evening Shift (Q#1)

An inductor of reactance 100Ω , a capacitor of reactance 50Ω , and a resistor of resistance 50Ω are connected in series with an AC source of $10V, 50Hz$. Average power dissipated by the circuit is _____ W.

Correct Answer: 1

Solution Analysis: To calculate the average power dissipated in the circuit, we follow these steps: Formula for Power : $P = V_{rms} \times I_{rms} \times \cos\phi$ Expressing Power in Terms of Voltage and Impedance : $P = V_{rms} \times \frac{V_{rms}}{Z} \times (R / Z)$ Simplifying the Expression : $P = \frac{V_{rms}^2 \times R}{Z^2}$ Calculate Impedance (Z) : The impedance of the circuit (Z) is calculated as: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ Given: - $X_L = 100\Omega$ - $X_C = 50\Omega$ - $R = 50\Omega$ $X_L - X_C = 100\Omega - 50\Omega = 50\Omega$ Thus: $Z = \sqrt{50^2 + 50^2} =$